

AI Chatbot Specification

■ Tuition Centres

AI chatbot specification for tuition centres, enrichment centres, and education businesses in Singapore.

Suggested Pricing

Component	Price
One-time setup fee	\$400 - \$600
Monthly maintenance	\$80 - \$120/month
Add-on: WhatsApp integration	+\$150-200
Add-on: Payment collection	+\$100-150
Add-on: Custom integrations	+\$100-200 each

Business Pain Points (What You Sell Against)

These are the problems the business owner already knows they have. Lead with these in your pitch — don't sell 'AI', sell the solution to their pain.

- Parents WhatsApp at 11pm asking about fees, schedules, and availability — no one is there to reply.
- Front desk spends 2+ hours/day answering the same 15 questions about class timing, pricing, and locations.
- Trial class bookings require manual back-and-forth on WhatsApp to find a suitable slot.
- Monthly fee collection is done manually — chasing parents on WhatsApp is awkward and time-consuming.
- Waitlist management is done on spreadsheets — parents aren't notified when slots open up.
- New parents visiting the website leave without enquiring because there's no instant response mechanism.

30-Second Sales Pitch

I'll put an AI assistant on your website and WhatsApp that answers parent enquiries 24/7 — fees, class schedules, trial bookings, everything. Parents get instant replies, you stop answering the same 15 questions every day. It takes 3 days to set up and costs less than one part-time admin staff per month.

Feature Requirements

Build in order: Must Have first (your MVP), then Should Have (first upsell), then Nice to Have (premium tier).

Feature	Description	Priority	Effort
FAQ from Documents	Answer questions about fees, subjects, schedules, policies from uploaded docs	Must Have	Low
Class Schedule Lookup	Find available classes by subject, level, and day of week	Must Have	Low
Trial Class Booking	Book a trial/assessment class by collecting student details and preferred slot	Must Have	Medium
Fee Enquiry	Detailed fee breakdown by subject, level, and package type	Must Have	Low
Waitlist Management	Auto-add to waitlist when class is full, notify when slot opens	Should Have	Medium
Monthly Fee Payment	Send payment reminders and collect fees via PayNow or Stripe	Should Have	Medium
Progress Update Routing	Parents ask about child's progress, bot routes to teacher	Nice to Have	Medium

Detailed Feature Specifications

Use these specifications when building with Claude Code. Each feature includes the user story, technical implementation notes, and acceptance criteria.

Feature: FAQ from Documents

Priority: Must Have | **Effort:** Low

User Story: As a parent, I want to ask about tuition fees and class schedules at any time and get an instant, accurate answer.

Technical Notes: RAG pipeline: chunk fee schedule PDF + class timetable + FAQ doc into vectors. Use Claude to generate answers grounded in retrieved context. If no relevant context found, respond with 'I don't have that information. Please contact us at [phone].'

Acceptance Criteria:

- Answers fee questions accurately within \$5 of actual price
- Handles variations: 'how much', 'what's the price', 'fees for', 'cost of'
- Returns 'I don't know' rather than hallucinating when info not in docs
- Response time under 3 seconds

Feature: Class Schedule Lookup

Priority: Must Have | **Effort:** Low

User Story: As a parent, I want to ask 'Is there Sec 3 Math on Saturdays?' and get the exact class time and availability.

Technical Notes: Store class timetable as structured data (JSON or CSV). Parse user query for subject, level, day. Return matching classes with timing, teacher, and remaining slots if available.

Acceptance Criteria:

- Correctly matches subject + level + day queries
- Shows class time, teacher name, and group size
- Handles abbreviations: 'Sec', 'Pri', 'JC', 'A-Math', 'E-Math'
- Handles day variations: 'Sat', 'Saturday', 'weekends'

Feature: Trial Class Booking

Priority: Must Have | **Effort:** Medium

User Story: As a parent, I want to book a free trial class for my child through the chatbot without calling the centre.

Technical Notes: Conversational flow: ask for student name, level, subject, preferred date/time. Check availability against Google Calendar API. Create calendar event and send confirmation. Email notification to centre admin.

Acceptance Criteria:

- Collects: student name, parent contact, student level, subject, preferred date/time
- Checks real-time availability via Google Calendar
- Creates confirmed booking with calendar invite
- Sends confirmation message with centre address and what to bring
- Notifies admin via email/Telegram when new booking created

Feature: Fee Enquiry

Priority: Must Have | **Effort:** Low

User Story: As a parent, I want to know exactly how much I'll pay per month, including any registration fees or material costs.

Technical Notes: Structure fee data as a lookup table: subject x level x class_type (group/individual). Include registration fee, material fee, and any sibling discounts. Return itemised breakdown.

Acceptance Criteria:

- Returns correct fee for any subject + level + type combination
- Mentions registration fee and material fee if applicable
- Mentions sibling discount and promo if active
- Handles 'is there a discount' or 'any promo' queries

Feature: Waitlist Management

Priority: Should Have | **Effort:** Medium

User Story: As a parent, I want to be notified when a slot opens up in a full class instead of having to keep checking.

Technical Notes: Store waitlist in a simple database (Supabase or Airtable). When a class cancellation occurs (manual trigger by admin), auto-notify first person on waitlist via WhatsApp/SMS. Waitlist position visible to parent.

Acceptance Criteria:

- Adds parent to waitlist with position number
- Notifies parent when slot becomes available
- Gives parent 24 hours to confirm before moving to next on list
- Admin can manually trigger slot opening

Feature: Monthly Fee Payment

Priority: Should Have | **Effort:** Medium

User Story: As the centre owner, I want parents to pay monthly fees on time without me having to chase them on WhatsApp.

Technical Notes: Generate PayNow QR code or Stripe payment link per student. Send automated reminder on 1st of month via WhatsApp. Track payment status. Alert admin for overdue payments after 7

days.

Acceptance Criteria:

- Sends payment reminder with correct amount and PayNow QR
- Tracks payment status: paid, pending, overdue
- Sends follow-up reminder after 3 days if unpaid
- Admin dashboard shows payment overview

Feature: Progress Update Routing

Priority: Nice to Have | **Effort:** Medium

User Story: As a parent, I want to ask about my child's progress and get connected to their teacher.

Technical Notes: Bot cannot answer progress questions (too specific). Instead, collect parent's query and student name, route to assigned teacher via email/Telegram. Teacher responds within the chat or separately.

Acceptance Criteria:

- Recognises progress-related queries and does NOT hallucinate answers
- Collects student name and parent's specific question
- Routes to correct teacher based on student's class
- Confirms to parent that teacher has been notified

Sample Conversations

These demonstrate expected chatbot behaviour. Use these as test cases during development.

Scenario: New parent asking about fees and booking a trial

Customer: Hi, do you have Sec 2 A-Math classes?

Bot: Yes! We offer Sec 2 A-Math every Tuesday 4-6pm and Saturday 10am-12pm. Group class (max 8 students) is \$280/month. Would you like to book a free trial class?

Customer: Saturday one, can try first?

Bot: Of course! I'll book you in for a free trial this Saturday 10am-12pm. Can I get your child's name and your contact number?

Customer: Ryan Tan, my number is 9123 4567

Bot: Booked! Ryan Tan for Sec 2 A-Math trial this Saturday 10am. Our centre is at Blk 123 Jurong East St 13, #02-45. Please bring a calculator and any recent exam papers. See you Saturday!

Scenario: Returning parent asking about adding a subject

Customer: My daughter is already taking English here. Want to add Science also. How much?

Bot: Great to hear! For existing students, Sec level Science group tuition is \$260/month. Since your daughter is already enrolled, there's no additional registration fee. We have Science on Wednesdays 6-8pm and Fridays 4-6pm. Which day works better?

Customer: Wednesday. Got sibling discount?

Bot: Yes! We offer a 10% sibling discount — so Science would be \$234/month instead of \$260. Shall I register your daughter for the Wednesday 6-8pm Science class?

Scenario: Class is full

Customer: Want to sign up for Pri 5 Math Saturday class

Bot: I'm sorry, our Saturday Pri 5 Math class is currently full (8/8 students). I can add you to our waitlist — you'll be first to know when a slot opens. We also have availability in the Thursday 4.30-6.30pm class. Would you prefer the waitlist or Thursday?

Technical Architecture

Core Stack (Required)

Component	Technology	Purpose
LLM	Claude API (claude-sonnet-4-20250514)	Understands questions and generates answers from context

Vector Database	ChromaDB (local) or Pinecone (cloud)	Stores document embeddings for RAG retrieval
Embedding Model	Voyage AI or OpenAI text-embedding-3-small	Converts text chunks to vectors
Backend	Python FastAPI	API server handling chat requests
Frontend Widget	Vanilla JS embeddable snippet	Chat bubble on client's website
Admin Dashboard	Streamlit or Next.js	Where business owner uploads docs and views analytics

Add-on Integrations

- **Google Calendar API:** Sync trial class bookings, manage class schedules and availability
- **WhatsApp Business API (via Twilio/360dialog):** Most SG parents prefer WhatsApp — critical for adoption
- **Stripe / PayNow:** Monthly fee collection, deposit for trial classes
- **Telegram Bot API:** Admin notifications — new bookings, payment alerts
- **Google Sheets / Airtable:** Simple backend for class data, waitlist, student records
- **SendGrid / Twilio SMS:** Email and SMS confirmations and reminders

Data Model (Documents to Collect from Client)

These are the documents and data you need from the business owner during onboarding. The quality of these documents directly determines chatbot accuracy.

- **Fee Schedule:** PDF or spreadsheet with all subjects, levels, and pricing (group vs individual, registration fee, material fee, discounts)
- **Class Timetable:** Spreadsheet: subject, level, day, time, teacher, max capacity, current enrollment
- **Centre Policies:** Withdrawal policy, makeup class policy, trial class terms, late payment policy
- **FAQ Document:** Top 20 most asked questions and answers (ask the front desk staff)
- **Centre Info:** Address, contact number, operating hours, MRT/bus directions, parking info
- **Teacher Profiles:** Teacher names, qualifications, subjects taught (optional, for trust-building)

PDPA Compliance Checklist (Singapore)

- Display clear data collection notice before collecting any personal data
- Get explicit consent before storing name, phone, email
- Provide opt-out option: 'Type STOP to opt out of data collection'
- Do not store NRIC, credit card numbers, or medical records in chat logs
- Implement data retention policy: auto-delete chat logs after 90 days

- Provide data export/deletion on request from the business owner
- Add privacy policy link in the chat widget footer

Singapore Localisation Notes

- Support Singlish phrasing: 'got discount ah?', 'how much one?', 'can book or not'
- Handle mixed language queries (English + Mandarin terms are common)
- Use SGD (\$) for all pricing — never USD
- Time format: 12-hour with AM/PM (e.g., 3pm, not 15:00)
- Support PayNow QR code generation for payments
- Reference local landmarks for directions ('near Jurong East MRT')
- Public holiday awareness for Singapore calendar